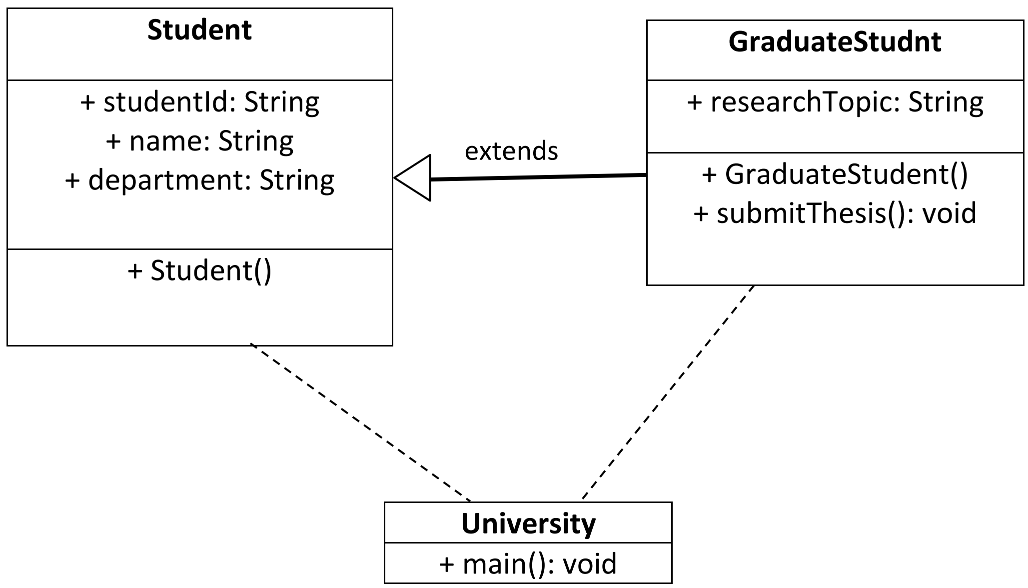
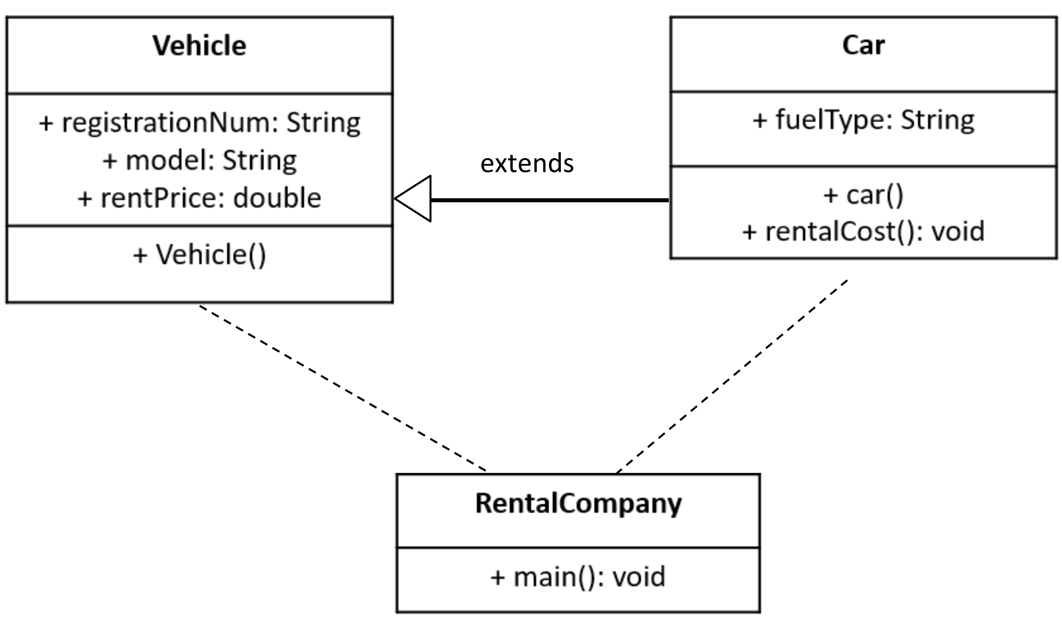
01. A university maintains student records using a Student class that contains studentID, name, and department. A GraduateStudent class inherits from Student and adds an attribute researchTopic along with a method submitThesis(), which prints a message confirming the thesis submission. Now Write a Java code to implement and execute the above scanrion.

UML diagram:



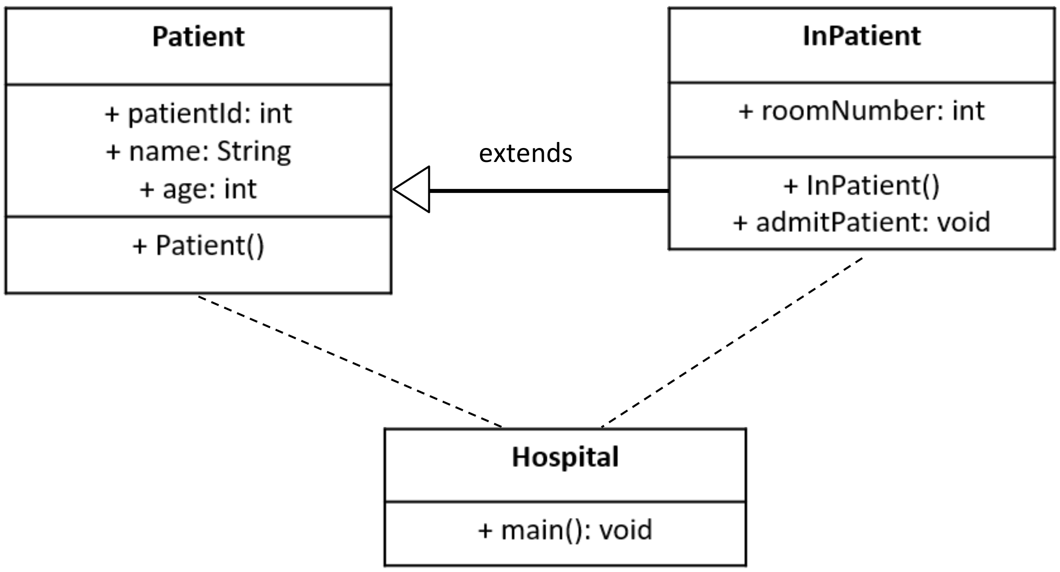
02. A vehicle rental company manages its fleet using a Vehicle class that includes registrationNumber, model, and rentalPrice. A Car class extends Vehicle and adds an attribute fuelType, along with a method calculateRentalCost(int days), which calculates the total rental cost based on the number of days the vehicle is rented. Now Write a Java code to implement and execute the above scanrion.

UML diagram:



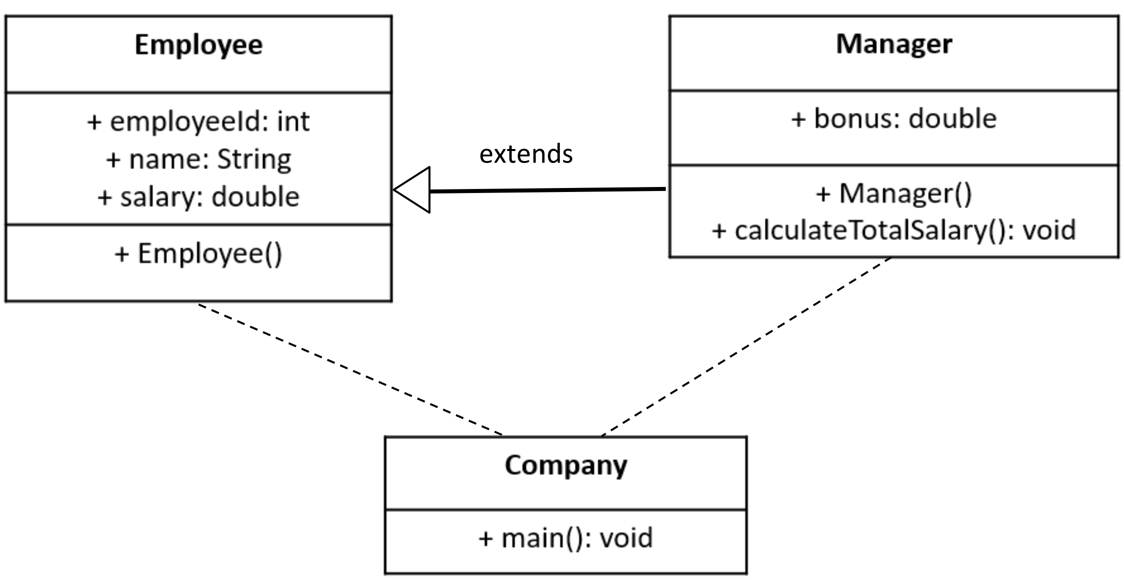
03. A hospital maintains patient records using a Patient class containing patientID, name, and age. A InPatient class extends Patient and adds an attribute roomNumber, along with a method admitPatient() that assigns a room and prints admission details. Now Write a java code to implement and execute the above scanrion.

<u>UML diagram:</u>



04. A company maintains employee records using an Employee class containing employeeID, name, and salary. A Manager class extends Employee and adds an attribute bonus, along with a method calculateTotalSalary(), which adds the bonus to the base salary and prints the total salary. Now Write a java code to implement and execute the above scanrion.

<u>UML diagram:</u>



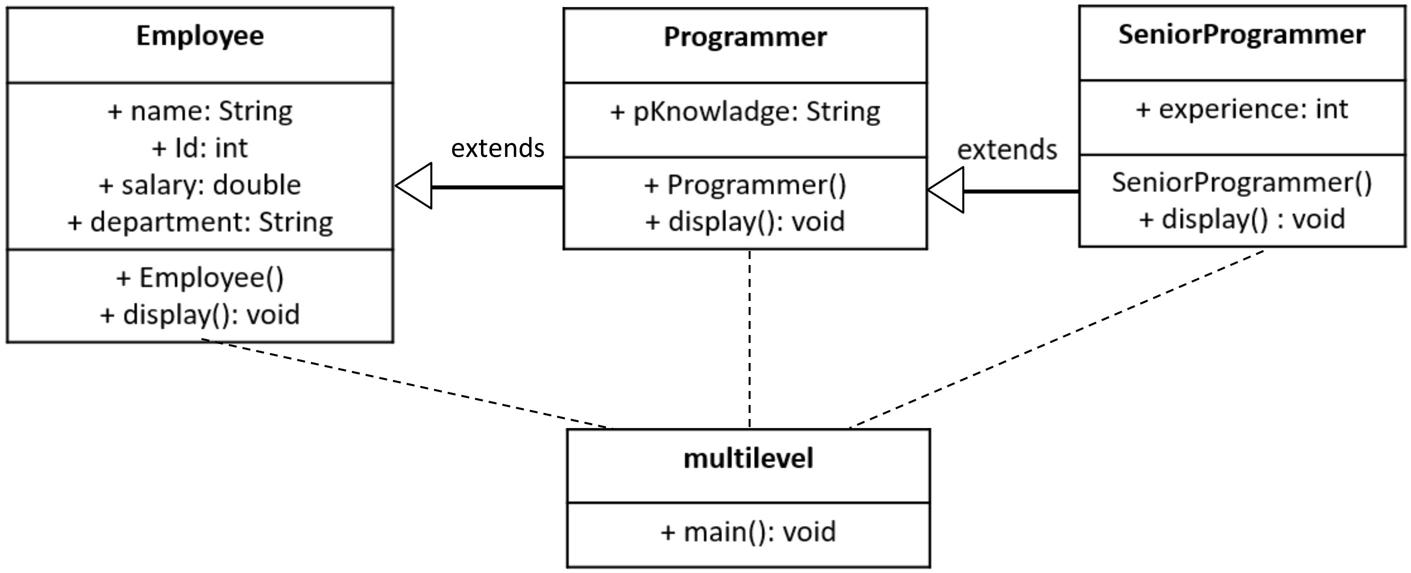
5. **UML for the multilevel inheritance code:**

Code:

```
class Employee {
	public String name;
	public int Id;
	public double salary;
	public String department;

	Employee(String name, int Id, double salary, String department) {
		this.name = name;
		this.Id = Id;
		this.salary = salary;
		this.department = department;
	}

	public void display() {
		System.out.println("Name: " + name + "\nId: " + Id + "\nDepartment: " + department +
		"\nSalary: " + salary);
	}

}

class Programmer extends Employee {
	public String pKnowladge;
	Programmer(String name, int Id, double salary, String department, String pKnowladge) {
		super(name, Id, salary, department);
		this.pKnowladge = pKnowladge;
	}
	@Override
```

```
    public void display() {
        super.display();
        System.out.println(pKnowladge);
    }
}

class SeniorProgrammer extends Programmer{
    public int experience;

    SeniorProgrammer(String name, int Id, double salary, String department, String pKnowladge, int
      experience){

    super(name, Id, salary, department,pKnowladge);
    this.experience=experience;
    }
    @Override
    public void display() {
        super.display();
        System.out.println(experience);
    }
}

public class multilevel{
    public static void main(String args[]){
    SeniorProgrammer sp1=new  SeniorProgrammer("sdf",101, 5000.56,"cse","C",5);
    sp1.display();
    }

}
```

UML diagram:


Employee
+ name: String
+ Id: int
+ salary: double
+ department: String
+ Employee()
+ display(): void
extends
Programmer
+ pKnowladge: String
+ Programmer()
+ display(): void
extends
SeniorProgrammer
+ experience: int
SeniorProgrammer()
+ display() : void
multilevel
+ main(): void